Token Arcade: The MiniCoin Economy

Pre-Whitepaper v0.2.2 - Codex-ready frozen economic spec
Founder, Metropolis judges/collaborators, and contracts build - 8 Oct 2026

> This document describes an experimental protocol architecture. It is not an offer or solicitation to purchase any token.
>
> **Source-of-Truth label:** Protocol / Economy.
>
> `v0` means it ships for Metropolis. `later` means it is specified but not built now. Numbers marked **(proposal)** are default configuration values, not hard-coded product promises. **(unverified)** marks a fact that must be checked before production use.
>
> If this document conflicts with older protocol/economy notes, this document wins. Product/application behavior is owned by the Token Arcade v0 Metropolis Handoff; protocol/economic behavior is owned here.

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1. Frozen mental model

Term	Meaning
Cabinet	A public game deployment in Token Arcade. It has its own `gameId`, leaderboard namespace, Raid history, community association, and optional economy.
Gameplay Skeleton	Reusable mechanics/simulation that may underpin multiple Cabinets. Skeleton reuse does not merge Cabinet leaderboards.
MiniCoin	Optional game economy: a game-specific ERC-20 plus its pool/configuration.
Community Asset	An existing community or asset a Cabinet supports: ERC-20, NFT collection, major asset, RWA concept, or other approved primitive.
Hook	`MiniCoinHook`. Captures programmable pool economics: charges the configured hook fee and records/attributes accrual. Nothing else.

Term	Meaning
Economy Router	`EconomyManager`. Allocates captured value across Strategies/destinations.
Strategy	Tells an allocation what to do: fund raids, support a community, go to treasury, later accumulate assets, etc.
Prize Vault	Holds free/locked Raid funding and manually deposited approved rewards for one MiniCoin.
Trigger	Decides when an Economy Raid can open. v0 uses `VaultThreshold` only.
RaidManager	Manages tournament economics: Raid records, escrow, settlement authorization, claims, expiry/cancel.
Score Verifier	Game-specific verification adapter. Deterministic replay where practical; authenticated session/telemetry or server-authoritative verification otherwise.
Passport	Player identity: Privy account plus embedded arcade wallet. Offchain in v0.

Cabinet identity invariant. A community may redeploy the same Gameplay Skeleton as a new Cabinet. That new deployment has a new `gameId`, leaderboard, Raid history, Community Asset association, and optional MiniCoin. Shared mechanics do not imply shared competitive history.

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2. What Token Arcade is

Token Arcade is a browser arcade for onchain communities, with Metropolis v0 launching on Monad.

A Cabinet can exist without a MiniCoin. A MiniCoin can be a standalone game economy, or it can be connected to an existing Community Asset through one or more configurable Strategies.

The MiniCoin itself is **not** the Strategy.

End-to-end value logic:

```
Hook captures configured pool fee
        ->
Economy Router allocates captured value
```

```
        ->
Strategies decide what allocations do
        ->
Raid/Reward configuration determines what players ultimately win
```

Principles:

- **No inflationary gameplay emissions.** MiniCoins are never minted as rewards. Rewards come only from assets already collected, acquired, or deposited.
- **No player-funded wagering in v0.** Metropolis Raids are free-entry and rewards are pre-funded. Paid Ranked entry remains later/legal-review scope.
- **No floor sweeping in the swap path.** v0 does not buy Community Assets during a swap.
- **No implied fund ownership.** MiniCoin holders do not automatically own or redeem Strategy/Vault/Treasury assets.
- **Game first, economy optional.** A Cabinet can launch now and add an approved MiniCoin later without changing its `gameId` or resetting its leaderboard history.

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3. Cabinets, modes, and what a MiniCoin adds

Mode	Needs MiniCoin?	Account	Money in (v0)	Rewards (v0)
Free Play	No	None/local	None	None
Ranked Run	No	Passport	None	Persistent leaderboard, cosmetics/status where implemented
Funded Raid	No	Passport	Free entry	Sponsor/operator/community-deposited ERC-20 and/or ERC-721 reward

Mode	Needs MiniCoin?	Account	Money in (v0)	Rewards (v0)
Economy Raid	Yes	Passport	Free entry	Prize escrowed from MiniCoin Prize Vault when trigger fires

A Community Raid always has: Cabinet/game, community, objective, start/end, entry/eligibility rules, attempts, leaderboard, reward definition, and settlement.

MiniCoin adds the autonomous economy layer: pool activity -> captured value -> configurable allocations -> Vault/Strategies -> economy-triggered events.

A Community Asset association can exist before a MiniCoin exists.

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4. The flywheel

```
Community / culture
      -> players + attention
Cabinet / game
      -> reason to participate / trade
MiniCoin / WMON pool
      -> hook fee
FeeAccumulator
      -> EconomyManager
      -> PrizeVault
      -> CommunityStrategy
      -> TA Treasury
PrizeVault threshold
      -> WOOP: Economy Raid opens
Raid
      -> prizes + status + community attention
      -> more play / activity
```

Non-ETF invariant. Assets held or accumulated by a Community Strategy are protocol/community-controlled treasury assets. MiniCoin ownership does not create a pro-rata ownership interest, redemption right, or claim on those assets. No contract may expose a function that redeems MiniCoins for Strategy, Vault, or Treasury assets.

5. The MiniCoin token

- Plain ERC-20.
- Fixed supply, minted once at launch to the Factory/launch flow.
- No mint function after deployment.
- No transfer tax.
- No blacklist.
- No owner-only pause.
- The token itself does not implement game rewards or Community Asset logic.

Why no transfer tax: v4 pool settlement expects exact token movement; fee-on-transfer behavior complicates/desynchronizes pool accounting. The economic fee belongs at the hook/pool layer, not inside the ERC-20.

Supply (proposal): `1,000,000,000` tokens, 18 decimals.

Allocation (proposal): founder-deployed v0 default is `0%` creator allocation and the remaining launch supply used for initial liquidity. Future creator allocation may be capped at `5%` and vested, but exact creator economics are not frozen for Metropolis.

Token Arcade does not require a `$TA` allocation. Protocol funding comes from the configured Treasury share.

6. Pool and pair asset

v0 pair asset: WMON only.

`WMON = 0x3bd359C1119dA7Da1D913D1C4D2B7c461115433A`

Direct MiniCoin/Community Asset pairs are an architectural possibility for later. They are not required for Metropolis.

PoolKey v0 (proposal):

- currencies: MiniCoin + WMON, sorted by address
- LP fee: `3000` = `0.30%` static
- tick spacing: `60`
- hook: current approved `MiniCoinHook` version
- dynamic-fee flag: **not used in v0**

The v4 LP fee and Token Arcade's custom hook fee are separate mechanisms.

A pool's hook is part of its PoolKey and cannot be replaced. A bad hook requires a new approved hook version, new pool, and controlled liquidity migration.

7. MiniCoinHook v1 - tiny, swap-critical

The hook should remain deliberately small.

Frozen economic responsibilities

1. Gate/validate approved pool initialization as required by the chosen Factory/deployment path.
2. Charge the configured custom hook fee on the WMON leg of a swap.
3. Attribute the captured WMON fee to the correct `PoolId` so it can later be distributed by `EconomyManager`.

v0 hook fee (proposal): `100 bps = 1.00%` of the WMON leg.
Hard ceiling: `500 bps = 5.00%`.

Per-pool hook fee is immutable for that pool in v0.

Swap-path rules

- only the PoolManager may invoke callbacks
- no Raid logic on the swap path
- no Community Asset purchase on the swap path
- no reads of Raid/Vault/Strategy state to decide whether swaps succeed
- no mutable admin function that changes an existing pool's hook behavior
- external failure in a downstream Strategy/Raid component must not brick the pool

Implementation hypothesis warning

The exact Uniswap v4 custom-accounting / PoolManager settlement calls used to capture/attribute WMON are an **implementation hypothesis, not a frozen Solidity API**.

Codex must validate the current installed Uniswap v4 interfaces and Monad deployment before implementing. Preserve the economic invariant - WMON hook fee is attributable by `PoolId` and later distributable - but refactor custody/accounting details if required to keep failure out of the swap-critical path.

Versioning

Each approved `MiniCoinHook` version may serve many pools. The Factory/Registry points new launches to the current approved version. Existing pools do not mutate to a new hook.

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8. FeeAccumulator, Economy Router, and Strategies

FeeAccumulator

Holds or accounts for captured WMON per pool with minimal logic. Only the approved EconomyManager path may distribute a pool's accrual.

EconomyManager / Economy Router

`distribute(poolId)` is intended to be permissionless where practical.

It allocates available WMON using that pool's configured basis-point split. Splits sum to 10,000.

v0 default allocation (proposal):

Allocation	Strategy / destination	Share
Raid funding	PrizeVault / RaidFundingStrategy	65%
Community	simple CommunityStrategy	20%
Token Arcade Treasury	TreasuryStrategy / multisig	15%
Creator	slot reserved; unused for founder v0	0%

Allocation changes, if supported, are delayed and evented. **Proposal:** 48-hour delay.

CommunityStrategy - simple v0

- Holds its Community allocation in WMON.
- Does not buy the Community Asset during a swap.
- In v0, may fund approved community/holder-specific Raids through the RaidManager path.
- Asset acquisition is a later `AssetAccumulationStrategy` with explicit slippage/min-out/per-call caps.

Strategy menu

Strategy	What it does	Status
RaidFundingStrategy	Allocation funds PrizeVault	v0
TreasuryStrategy	Allocation to TA Treasury	v0

Strategy	What it does	Status
CommunityStrategy	Holds WMON / funds approved community Raid use	v0 simple
VaultThreshold Trigger	Opens Economy Raid when free Vault balance reaches threshold	v0
HolderRewardStrategy	Rewards verified Community Asset holders	later
AssetAccumulationStrategy	Swaps allocated WMON into approved Community Asset asynchronously	later
DynamicFeeStrategy	Different/new hook version with dynamic fee logic	later
LPStrategy	Deepens/manages liquidity from an allocation	later
Time / Volume / Holder / Composite triggers	Other event conditions	later

The creator-facing launch UI may eventually expose these as safe audited toggles. v0 implements only the minimal modules needed to prove the loop.

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9. PrizeVault

- One PrizeVault per MiniCoin, preferably minimal clone/proxy pattern if appropriate after security review.
- Holds WMON from RaidFundingStrategy.
- Can also hold manually deposited approved ERC-20 and ERC-721 rewards.
- Tracks free vs Raid-locked balances.
- Pays only what it holds; never mints.
- No arbitrary admin `withdraw(address)`.
- Funds leave through approved Raid escrow/claim/cancel/migration paths only.

When an Economy Raid opens, PrizeVault reserves the Raid's prize from free balance.

Per-Raid prize cap (proposal): at most 25% of free WMON balance, additionally capped by a per-Cabinet `maxPrize`.

Unclaimed rewards return according to Raid source after the configured claim window.

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10. Triggers, Raids, verification, settlement, claims

v0 trigger: VaultThreshold only

`openEconomyRaid(gameId)` can succeed when:

- the MiniCoin PrizeVault free balance >= configured threshold
- no conflicting Economy Raid is active for that Cabinet
- configured cooldown has passed
- the Cabinet/Verifier/economy configuration is still valid

When it opens:

- WMON prize is reserved from the Vault
- optional pre-approved/manual NFT hero reward may also be queued
- Raid start/end/rules are recorded
- `RaidOpened(...)` is emitted

Why VaultThreshold only in v0:

- objective and onchain
- cheap to evaluate
- forcing the trigger by trading also pays the fees funding the Raid
- volume triggers invite wash-trading design problems
- market-cap triggers need oracle/TWAP design
- holder count/concurrent-player triggers need offchain/indexed inputs

Funded Raids

A Cabinet does not need a MiniCoin to run a Funded Raid. An operator/community/sponsor may deposit approved rewards and create a Raid through the same settlement layer.

Score Verifier

The browser is never the final authority.

Each Cabinet declares one verifier approach:

- **session/telemetry validation** - authenticated run token plus game-specific checks/plausibility/telemetry

- **server-authoritative verification** - trusted server/judge owns hidden/authoritative state
- **deterministic replay** - replay exact version/seed/input log and recompute result

Deterministic replay is preferred where practical, but **not mandatory for every Raid-ready game**.

Every verified run must be bound to at least:

- `gameId`
- game/simulation version
- mode/Raid id
- rules/rulesHash or equivalent
- run/session id
- player identity/account
- accepted metrics/result

Replay protection and tamper rejection are mandatory.

v0 settlement authorization

The v0 settlement signer is centralized and bounded to the escrow of the specific Raid being settled.

For a Raid with WMON plus an optional hero NFT, use explicit reward semantics rather than overloading one generic amount field.

A practical v0 settlement shape may be conceptually:

```
settle(
  raidId,
  erc20MerkleRoot,
  totalErc20Payout,
  nftWinner?
)
```

The Raid record already knows the escrowed ERC-20 asset/amount and optional NFT contract/tokenId.

Players claim ERC-20 distributions with proofs to their Passport embedded wallet. The optional hero NFT is transferred only to the recorded winner through the Raid settlement/claim path.

No settlement may distribute more than the Raid's escrow.

Unawarded/unclaimed assets return to the appropriate source after configured windows. If settlement never arrives, a timeout/cancel path returns escrow to source.

Proposal: claim window 14 days; settlement timeout 7 days.

Trust path forward

- publish useful verification artifacts/logs where practical
- add challenge windows
- stronger server attestation / TEE where useful
- zk verification for games where cost/complexity is justified

Privy Passport reduces casual multi-wallet duplication; it is not Sybil-proof.

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11. Launch structure - MiniCoinFactory

The product presents a **single user-facing launch flow; atomic where practical**. Do not assume every step fits safely in one transaction until deploy/approval/liquidity mechanics are tested.

Conceptual steps:

1. Deploy fixed-supply MiniCoin.
2. Initialize approved v4 MiniCoin/WMON pool with the current approved Hook version.
3. Seed liquidity through PositionManager using an explicitly validated launch config.
4. Lock/custody the LP position according to the approved migration policy.
5. Create/wire PrizeVault, CommunityStrategy, EconomyManager/Raid config.
6. Register `gameId -> {MiniCoin, poolId, hookVersion, vault, communityStrategy, communityAsset}`.
7. Existing Cabinet flips from game-only to MiniCoin-enabled without changing `gameId` or leaderboard history.

Creator/Admin product lifecycle vs protocol authority

Product v0 uses:

```
SUBMIT -> ADMIN APPROVE -> DEVELOPER DEPLOY -> PUBLISHED
```

The **developer initiates Deploy after approval**.

This does **not** require giving every creator unrestricted Factory authority in Metropolis v0. The approved Deploy action may invoke a platform-controlled/allowlisted Factory path, Registry approval, signed launch authorization, or equivalent constrained mechanism.

For the demo, the founder may act as both creator and admin.

Future v1 expands approved creator launch authorization after audit. Future permissionless deployment must remain constrained to audited templates/Strategies/guardrails.

Liquidity mechanics are not frozen

The following remain explicit Open Questions and must be parameterized/tested:

- single-sided vs two-sided seed liquidity
- start price
- range
- LP lock duration/permanence
- exact migration mechanism

The current single-sided/full-supply concept is a candidate, not economic truth.

Do not deploy irreversible mainnet contracts or seed real mainnet liquidity without explicit founder approval after these parameters are resolved.

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12. Contract picture - v0

```
MiniCoinFactory
  |-- MiniCoin
  |-- v4 Pool -- MiniCoinHook v1 -- FeeAccumulator -- EconomyManager
  |                                                   |-- PrizeVault
  |                                                   |-- CommunityStrategy
  |                                                   `-- TA Treasury
  |-- Registry
  `-- LP Lock

PrizeVault -- VaultThreshold --> RaidManager
                                 |-- escrow
                                 |-- RaidOpened event
                                 |-- settlement authorization
                                 `-- claims / sweep / cancel
```

Contract	Responsibility
`MiniCoinFactory`	Approved launch flow; enforces caps/approved Hook version/config boundaries
`MiniCoin`	Plain fixed-supply ERC-20
`MiniCoinHook v1`	Gate approved pool init as required; capture static WMON hook fee; attribute accrual
`FeeAccumulator`	Minimal per-pool WMON accrual custody/accounting

Contract	Responsibility
`EconomyManager`	Per-pool allocation and delayed config changes
`PrizeVault`	Free/locked balances; WMON + manually deposited approved ERC-20/ERC-721 rewards
`CommunityStrategy`	v0 simple WMON community allocation / approved Raid funding
`RaidManager`	VaultThreshold Economy Raids, Funded Raids, escrow, settlement, claims, expiry/cancel
`Registry`	Cabinets/games, Community Assets, optional MiniCoin/economy refs, approved Hook versions, creator/submission state as needed
`LP Lock`	Holds LP position; allows only approved recovery/migration path
`TA Treasury`	Multisig wallet, not custom token logic

Player identity, scores, cosmetics, and run artifacts remain offchain in v0, keyed to the Token Arcade/Privy identity model.

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13. Monad / Uniswap v0 references

Monad mainnet chain id: `143`.

Current references from the prior verified spec; **re-check all addresses/interfaces immediately before deployment**:

- **PoolManager** - `0x188d586ddcf52439676ca21a244753fa19f9ea8e`
  Previously verified against the official Uniswap SDK.
- **PositionManager** - `0x5b7ec4a94ff9bedb700fb82ab09d5846972f4016`
  Previously verified against the official Uniswap SDK.
- **WMON** - `0x3bd359C1119dA7Da1D913D1C4D2B7c461115433A`
  Canonical reference in the current notes.
- **Universal Router v2.1.2** - `0xa6CE4F10d83dBdDAc17E68e1837ca9cE6a1b596e`
  Current target in the existing spec.
- **Older Universal Router** - `0x0d97dc33264bfc1c226207428a79b26757fb9dc3`
  Older deployment; do not target by default.

- **Permit2** - `0x000000000022D473030F116dDEE9F6B43aC78BA3`
  Re-verify before v0 deployment.

Routing wording: v0 does not depend on third-party frontends automatically discovering the MiniCoin pool. Token Arcade should execute supported swaps through the chosen current router/path using the explicit approved pool configuration.

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14. Metropolis Cabinet / economy map

The initial Cartridge integration and the economic demo are different steps.

eMONAD Cabinet

- **Community concept:** eMONAD
- **Founder-supplied reference:** `0x81a224f8a62f52bde942dbf23a56df77a10b7777`
- **Game/theme state:** built from the Buy Bounce source/Skeleton; public branding becomes eMONAD-specific.
- **Economy state:** a game-specific MiniCoin is planned/compatible, but it is not implemented inside the Cartridge-integration run. Verify current Community Asset/address/liquidity before live protocol deployment.

Two Places at Once

- **Community concept:** Skrumpeys ERC-721
- **Game/theme state:** preserve the existing game. Additive Skrumpey enemies/traits/content may come later.
- **Economy state:** Community association can exist before the MiniCoin. A game-specific MiniCoin/Community Strategy may be enabled later without changing the Cabinet ID. No fungible Skrumpeys pair is implied.

Coindle

- **Community concept:** COIN / RWA
- **Game/theme state:** no reskin.
- **Economy state:** **Game-only / Economy Coming Soon** until an approved supported asset path exists.

Skip to 10K

- **Community concept:** ETH / WETH
- **Game/theme state:** no reskin.
- **Economy state:** game-only/economy-compatible metadata for now. Do not deploy against an unverified bridged WETH path. Stretch second

MiniCoin only after infrastructure is verified and the primary economic loop is stable.

Skrumpey Slide

- **Community concept:** Skrumpeys ERC-721
- **Game/theme state:** intentionally withheld from the initial legacy import; it enters through the real Creator Portal.
- **Economy state:** hero end-to-end economic target after `Submit -> Approve -> Developer Deploy`; MiniCoin/WMON -> Hook -> PrizeVault -> VaultThreshold -> Raid.

No partnership with any referenced community is implied unless separately confirmed. Use stand-ins where rights/permission are not available.

Skrumpey NFT reward v0: manual approved NFT deposit. Automated NFT acquisition is later. When an approved NFT is present in the Raid reward configuration, the Raid can settle it to the winner without the game itself holding NFT logic.

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15. v0 parameters - proposals unless stated

Parameter	v0 value	Mutable?
Pair asset	WMON	fixed for Metropolis v0
LP fee	0.30% static	PoolKey / immutable for pool
Hook fee	1.00% WMON leg; max 5%	immutable per pool in v0
Allocation	65 / 20 / 15 / 0	delayed config where supported
Supply / founder creator allocation	1B / 0% default	fixed at launch
Trigger	VaultThreshold	v0 decision
`prizeBps`	2500 = 25% proposal	per future Raid config, subject to maxPrize
Claim window	14 days proposal	future Raids only
Settlement timeout	7 days proposal	future Raids only

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16. Roadmap - protocol/economy

- **v0 Metropolis:** curated MiniCoins, WMON pair, static Hook fee, FeeAccumulator, EconomyManager, simple CommunityStrategy, PrizeVault, VaultThreshold Economy Raids, free-entry Funded/Economy Raids.
- **v1:** approved/audited creator launch authorizations; safer Strategy toggles at launch; stronger operational controls/audit.
- **v2:** AI/Mad-Lib creator launchpad against audited Gameplay Skeletons/Cartridges and constrained economic templates.
- **later:** direct Community Asset pairs, DynamicFeeStrategy/new Hook versions, AssetAccumulationStrategy, HolderRewardStrategy, LP-Strategy, more TriggerStrategies, automated NFT acquisition after marketplace/security design, broader multichain adapters.
- **not v0:** betting/wagering/paid-entry pots; automated floor sweeps; `$TA` dependency.

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17. Risks and guardrails

Hook / pool risk

- Keep Hook tiny.
- Treat current v4 call details as implementation hypotheses until compiled/tested against current dependencies.
- Fuzz conservation: captured fee == attributed/distributable fee.
- Do not let Raid/Strategy failure become swap failure.
- Hook cannot be patched in place; recovery means new Hook version/pool + controlled migration.

Vault / Raid risk

- no arbitrary admin withdrawal
- signer bounded to Raid escrow
- mixed ERC-20/ERC-721 settlement explicit
- cancellation/timeout returns assets to source
- hostile tests before creator launch expansion

Score cheating

- client untrusted
- game-specific Score Verifier
- session/rules/version binding
- replay protection
- deterministic replay only where practical, not a blanket rewrite requirement

Liquidity / launch risk

- seed/range/start-price/lock mechanics not yet frozen
- no irreversible mainnet liquidity action without founder approval
- creator allocation/caps enforced by approved Factory config when introduced

Community/IP relations

- do not claim partnership without permission
- NFT/brand asset availability does not automatically equal commercial rehosting/use rights
- stand-ins are labeled

Regulation / wagering

- v0 Raids are free entry
- no player-funded pot
- no yield/revenue-share/price-promise language
- non-ETF invariant: no redemption path into Strategy/Vault/Treasury assets
- RWA/tokenized-equity concepts remain game-only until compliant supported assets are verified

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18. Open questions

1. Final seed-liquidity structure, start price/range, LP lock duration, and migration mechanism.
2. Final Hook/LP fee values and 65/20/15 split; config-delay length.
3. Per-Cabinet Raid threshold, `prizeBps`, `maxPrize`, cooldown, duration, attempts, tie-breaks.
4. Which Score Verifier each Cabinet uses in Metropolis; whether any verifier warrants a challenge window now.
5. NFT/community holder eligibility: live balance, snapshot, delegated ownership, and timing of checks.
6. Skrumpeys permission / hero NFT source; stand-in if unavailable.
7. TA Treasury multisig signers/spending policy.
8. Gas sponsorship: which actions, if any, are sponsored on mainnet.
9. When CommunityStrategy funding becomes rule-based instead of operator-triggered.
10. Exact creator-facing Deploy authorization: Registry allowlist, signed launch authorization, relayer, or other constrained v0 mechanism.
11. Verify the current eMONAD Community Asset/address/liquidity before economic use.

12. Verify the current Monad WETH/bridged ETH path before enabling Skip to 10K economy.

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19. Changelog - v0.2.1 -> v0.2.2

- Added **Cabinet vs Gameplay Skeleton** identity rule; leaderboards are per Cabinet/deployment, not per Skeleton.
- Aligned creator lifecycle to **Submit -> Admin Approve -> Developer Deploy -> Published**, while preserving constrained v0 protocol authority.
- Updated Metropolis Cabinet map:
  - Buy Bounce source becomes an eMONAD-branded Cabinet rather than a public "Buy Bounce" listing.
  - Two Places is associated with Skrumpeys without a broad reskin.
  - Coindle stays game-first with COIN/RWA economy unavailable/Coming Soon.
  - Skip to 10K stays unreskinned with ETH/WETH concept and no unverified WETH deployment.
  - Skrumpey Slide is withheld from initial import and becomes the hero real submission/deployment path.
- Reaffirmed game-specific **Score Verifier** model; deterministic replay is not universally required.
- Clarified mixed WMON + optional hero NFT settlement.
- Preserved Hook -> FeeAccumulator -> EconomyManager -> Strategies separation.
- Preserved VaultThreshold as the only v0 automatic trigger.
- Reaffirmed unresolved liquidity mechanics as parameters/open questions rather than defaults Codex may invent.

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20. Codex implementation directive

This document is the Protocol/Economy Source of Truth.

Do not redesign its economics during implementation.

Before writing production Solidity:

1. inspect the repository's current Uniswap v4 dependencies
2. verify current Monad deployment addresses/interfaces
3. compile a minimal spike for the Hook custom-accounting path
4. flag API mismatches rather than silently changing the economic model
5. write unit/fuzz/invariant tests before irreversible deployment
6. do not seed real mainnet liquidity without explicit founder approval

The smallest required protocol slice is:

```
MiniCoin
-> Pool
-> Hook fee
-> per-PoolId accrual
-> EconomyManager
-> PrizeVault
-> VaultThreshold
-> RaidOpened
```

Then connect the already-built application/cartridge layer to that slice.